Vishesh Shekhawat

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Education

M.Tech in Artificial Intelligence (Generative AI) — Aug 2025 – Present
NIT Jalandhar — **CGPA: 8.78/10**

B.Tech in Computer Science & Engineering — 2020 – 2024
Sobhasaria Group of Institute, Sikar — **CGPA: 8.45/10**

Projects

Modular Adaptive Network OS (manOS) — *FastAPI, PostgreSQL, FAISS, RAG, PyTorch*

- Designed modular AI framework for adaptive learning, isolated memory orchestration, and scalable multi-agent knowledge systems
- Built end-to-end pipeline covering ingestion, semantic chunking, retrieval augmentation, LLM-based QA generation, and vector storage integration
- Developed baseline research workflows for adaptive QA-pair generation and iterative knowledge refinement
- Integrated FAISS-based retrieval enabling context-aware generation across evolving knowledge sources
- Implemented modular FastAPI backend using repository patterns for scalable orchestration and experimentation

NewStack-AI – Modular LLM News Intelligence Pipeline — *Python, LLMs, RAG*

- Built modular pipeline aggregating AI-related content from YouTube transcripts and RSS feeds
- Transformed unstructured HTML and transcript data into structured markdown using LLM-based summarization workflows
- Implemented ranking and personalization logic to score relevance and reduce duplicate information delivery
- Integrated retrieval-grounded metadata filtering to improve factual consistency and reduce hallucinations
- Automated personalized HTML digest generation with centralized orchestration pipelines

Diffusion Models & Stable Diffusion Experiments — *PyTorch, CLIP, DDPM*

- Implemented Stable Diffusion pipeline for text-to-image generation tasks
- Explored scheduler optimization, classifier-free guidance, latent manipulation, and prompt conditioning techniques

Hackathons & Competitions

StatHack – Skin Disease Classification (IIT Goa) — *Computer Vision*

- Built CNN-based medical image classification pipelines with preprocessing and augmentation techniques
- Achieved **6th rank** on Kaggle evaluation leaderboard among competitive participants

Heart Disease Prediction – ML Competition — *Machine Learning, Periodic-Encoding*

- Built ensemble models on 600k+ records achieving ROC-AUC score of 0.955 through feature engineering, ensembling and fine-tuning
- Ranked in **Top 5.4% (238 / 4370)** demonstrating strong predictive modeling performance

Achievements

- Achieved **Rank 137** on TensorTonic Platform.
- Achieved **Rank 1327** on deep-ml Platform.
- Solved **300+ DSA problems** on LeetCode (Top 32%).
- Qualified **GATE 2024 & GATE 2025**.

Skills & Interests

Programming: Python, C/C++, SQL
Frameworks & Tools: PyTorch, Scikit-learn, FastAPI, PostgreSQL, FAISS
AI/ML: Generative AI, RAG, LLMs, Diffusion Models, Prompt Engineering, Pipeline Design, REST APIs
Research Interests: Generative AI, QA Pair Generation, Diffusion Models, Stable Diffusion, GAN, VAE.